

CHILDHOOD INFECTIONS

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Childhood exanthems : Infectious diseases characterized by rash.

Classification

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- Arranged in order of historical importance.

Number	Disease
1st disease	measles
2nd disease	Scarlet Fever
3rd disease	Rubella
5th disease	Erythema Infectiosum
6th disease	Roseola infantum/exanthema subitum

Measles

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Measles virus (ss RNA virus).

Incubation period : 8 - 12 days.

mode of spread : Respiratory droplets.

Secondary attack rate : >90% (Highly infectious/contagious).

Infectivity period/period of isolation : 4 days before to 5 days after onset of rash.

CLINICAL FEATURES :

1. Prodromal stage : Day 1-3
 - Fever (Intensity ↑ after rash appears).
2. Exanthem : Day 2-3 (Internal manifestation).

- 3C's.
 - Cough.
 - Coryza.
 - Conjunctivitis.



Koplik spot : Characteristic

- white spot surrounded by reddish hue (usually opposite to and lower molar)



Rice grain/grain of sand appearance

3. Exanthematous phase : Day 4-8 (External manifestation).

- maculopapular rash :
 - Site : Behind ear/face
 - ↓ Downward spread
 - Trunk and extremities
 - Leaves behind brownish pigmentation
 - ↑ fever intensity after rash appears.



measles rash

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COMPLICATIONS :

Acute :

- **most common** : Otitis media.
- **most severe** : Pneumonia (↑est mortality risk)
 - Due to virus : Giant cell/**HECHT** pneumonia
 - Due to superimposed bacterial infection (Staph. aureus/Strep. pneumoniae).

Chronic : **SSPE** (Subacute sclerosing pan encephalitis).

- Rare (1 in 1,00,000 cases).
- Latent complication : 7-13 years after primary infection.
- Clinical features :
 - **myoclonic seizures**.
 - Cognitive impairment (H/o recent deterioration in school performance).
 - Personality changes.
- Diagnosis : IgG anti-measles antibody present in CSF.
- Fatal condition despite immunomodulatory treatment.

MANAGEMENT :

- Supportive treatment : Adequate nutrition, antipyretic and IV fluids.
- **Vitamin A** supplementation : ↓es morbidity and mortality by 50%.
 - 2 doses 24 hours apart.
 - Dose
 - >1 year old : 1 lakh IU
 - <1 year old : 2 lakh IU

Scarlet Fever

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Etiology :

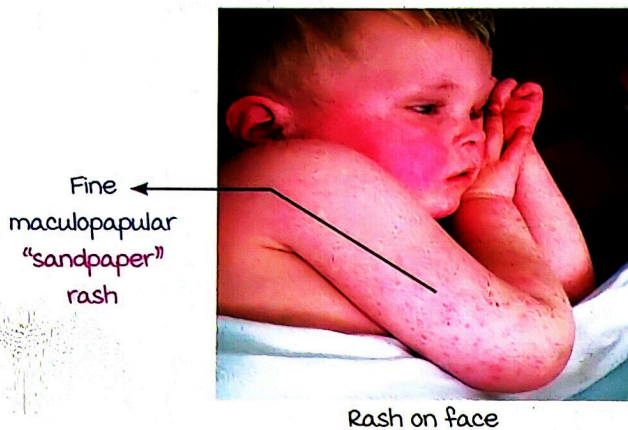
- **Group A β hemolytic streptococci** → Toxin : Pyrogenic exotoxin (Erythrogenic toxin)

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Age group : 5 - 15 years.

CLINICAL FEATURES :

- Fever.
- Sore throat.
- Tender anterior, cervical lymphadenopathy.
- **Strawberry tongue** (D/t hypertrophy of papillae).
- Rash : Starts on and day of fever.
 - **Spare palms and soles**

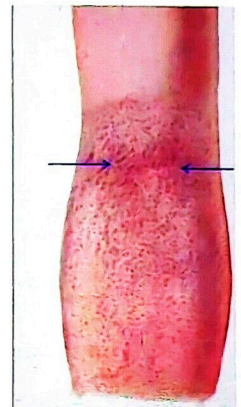


Rash on face



Trunk and extremities

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Pastia's lines :
Rash accentuation at
skin creases**POST STREPTOCOCCAL COMPLICATIONS :**

1. Acute rheumatic fever.
2. Post streptococcal glomerulonephritis (PSGN).
3. Post streptococcal reactive arthritis.

MANAGEMENT :

Investigations : Throat swab culture.

Treatment : Oral Penicillin V for 10 days > Amoxicillin for 10 days.

Rubella

00:22:34

- AKA German measles/3 day measles (Similarity in C/F's).

Etiology : *Togaviridae* (ss RNA virus).

Incubation period : 14-21 days.

mode of spread : Respiratory droplets.

Infectivity period : 5 days before - 6 days after onset of rash.

CLINICAL FEATURES :**1. Rash :**

- First sign/starts on Day 1 of illness (without prodromal stage).
- maculopapular rash.
- Starts from behind the ear/face → Spreads downwards.
- No desquamation/dicoloration after rash disappears.



Rubella rash

2. Exanthem :

Förchheimer spots : Tiny, rose lesions in oropharynx.

Lymphadenopathy : Suboccipital, posterior auricular and anterior cervical lymph nodes.

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COMPLICATIONS :

1. Post infectious thrombocytopenia:

- 2-3 weeks after infection.
- Self-limiting condition.

2. Arthritis :

- Seen in older children.

3. Progressive Rubella Panencephalitis (PRP) :

- **most severe.**
- Features similar to SSPE.

Erythema infectiosum

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Etiology : Parvovirus B19 (ss DNA virus).

Incubation period : Between 4-28 days.

Age group : >5 years.

CLINICAL FEATURES :

1. Prodromal stage :

- Fever for 1-2 days.

Exanthematous phase (External manifestation) :

Stage 1 : **Slapped-cheek** appearance,
facial flushing.



Slapped cheek appearance

Stage 2 : Diffuse macular erythema,
(Spread of rash to trunk & extremities).

Stage 3 : Central clearing of rash,
lacy/reticulated appearance.



Central clearing of rash

ASSOCIATIONS :

Arthralgia :

- In older children.

Transient aplastic crisis :

- Pre-existing hemolytic anemia + parvovirus B19 infection.
- Presents with pancytopenia.

Infection in the mother (during pregnancy).

MANAGEMENT :

- Supportive management (with good prognosis).

Roseola infantum/Exanthema subitum

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Etiology : Human herpes virus (HHV) 6 > 7, echovirus 16.

Age group : 6 months – 3 years.

Clinical features :

I. Prodromal stage (1-4 days) :

- Fever.
- Cough.
- Running nose.

II. Exanthematous phase (On day 4) :

- maculopapular rash.
- Starts in the trunk.
- ↓ fever intensity after rash appears.
- No residual pigmentation.



Rashes over trunk

III. Exanthem :

Nagayama spots : Erythematous macules in soft palate and uvula.

Treatment :

Prognosis :

- Supportive treatment.

- Good.

HFMD (Hand Foot Mouth Disease (HFMD))

00:39:46

Age group : Preschool age (< 5 years).

Causative agent : Coxsackie A16, Enterovirus 71.

mode of spread : Direct contact/infected fomites
(Eg : used towel).

Clinical features :

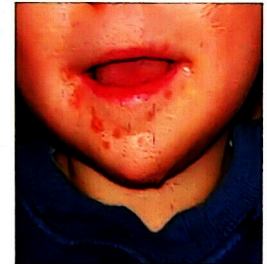
- Fever.
- Vesicles : Resolves over 4-5 days.

Treatment : Supportive.

m/c Complication : Temporary loss of nails 4 weeks after initial infection.



Papulovesicular lesions
in soles and palms



Blister around oral cavity
(First manifestation)

Varicella/Chicken pox

00:43:40

Characteristics :

Causative agent : Varicella-zoster virus (ds DNA virus).

Incubation period : 10-21 days.

Mode of spread : Droplet spread.

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Infective period/isolation period : 24-48 hours before onset of rash
→ crusting of all lesions

Secondary attack rate : 80%.

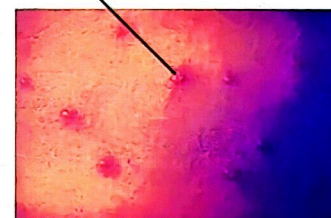
Clinical Features :

- Fever.
- Rash : Day 2 of fever.
 - Lasts for 3-5 days.
 - macule → Papule →
 - Vesicle** (Classical rash) : Pruritic lesion
 - Starts in trunk → Spreads to other areas.
 - Leaves behind hypo/hyperpigmented macules.
 - **Pleomorphic** appearance.

'Dew drop on a rose petal'
appearance



Varicella rash



Fluid filled vesicles

Complications :

- Secondary bacterial infection of skin lesion
 - Staph. aureus
 - Strep. pyogenes
- Neurological : **Cerebellar ataxia**, meningo-encephalitis.
- Reye's syndrome (Aspirin usage in varicella).
 - Hepatic encephalopathy.

Diagnosis :

- Primarily clinical.
- Lesion scrapped → **Tzanck smear** → multinucleated giant cells (Poor sensitivity).
- Serology : Anti-IgM antibody against varicella.

Treatment :

- Antipyretics : Paracetamol.
- Antiviral :
 - Acyclovir : 20mg/kg/dose x 4 times/day.
 - Start **within 48 hours**.
- **Avoid Ibuprofen** (↑ risk of necrotizing fasciitis).

Prevention : If not administered previously → Post exposure prophylaxis :

- Varicella vaccine.
 - Live attenuated **Oka strain**.
 - Dose : 2 doses SC.
 - 1st dose : 15-18 months
 - 2nd dose : 4-6 years
- Varicella vaccine.
 - **VZIG** (High risk contacts) :
 - Immunocompromised children.
 - Pregnant women.
 - Neonate born to mother who was affected (5 days before to 2 days after delivery).